

BRIGHTLEARN

PRACTICAL 4: MICROSOFT SQL SERVER FUNDAMENTALS

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QUESTION 1:

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37
38 -- 1. View the first 100 rows of the dataset to understand its structure.
39 SELECT TOP (100) *
40 FROM [SQLFundamentals].[dbo].[ketrosoccerdatasetadvanced];
41
```

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	player_name	team	date_of_birth	age	marital_status	number_of_kids	nationality	country_of_birth	position	preferred_foot	height_cm	weight_kg	jersey_number	injury_status
1	Siyanda Dlamini	Stellenbosch FC	1995-05-28 00:00:00.0000000	30	Widowed	0	South African	South African	Defender	Right	167	71	35	Recovered
2	Thabo Ndlovu	Cape Town City	2004-06-01 00:00:00.0000000	21	Single	1	Zimbabwean	Zimbabwean	Forward	Right	183	73	40	Recovered
3	Vusi Molefe	Stellenbosch FC	2006-09-20 00:00:00.0000000	19	Single	0	Nigerian	Nigerian	Forward	Left	167	66	65	Recovered
4	Thembi Mahlangu	Bloemfontein Celtic	2004-02-08 00:00:00.0000000	21	Divorced	0	Zambian	Zambian	Goalkeeper	Left	166	93	24	Injured
5	Nokuthula Sithole	Polokwane City	2003-03-18 00:00:00.0000000	22	Divorced	3	Nigerian	Nigerian	Goalkeeper	Both	193	88	79	Injured
6	Thembi Sithole	Kaizer Chiefs	1989-02-14 00:00:00.0000000	36	Married	2	Nigerian	Nigerian	Goalkeeper	Left	193	79	54	Health
7	Siyanda Mahlangu	Chippa United	1989-12-09 00:00:00.0000000	36	Single	0	Zambian	Zambian	Defender	Both	181	67	27	Health
8	Lerato Mashaba	Polokwane City	1998-01-31 00:00:00.0000000	27	Widowed	0	Ghanaian	Ghanaian	Forward	Right	181	65	85	Health

QUESTION 2:

```
40
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42 -- 2. Count the total number of players in the dataset.
43 SELECT COUNT(DISTINCT player_name) AS Number_of_Players
44 FROM [SQLFundamentals].[dbo].[ketrosoccerdatasetadvanced];
45
```

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	Number_of_Players
1	203

QUESTION 3:

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47 -- 3. List all unique teams in the league.
48 SELECT DISTINCT(Team)
49 FROM [SQLFundamentals].[dbo].[ketrosoccerdatasetadvanced];
50
51
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	Team
1	AmaZulu FC
2	Bloemfontein Celtic
3	Cape Town City
4	Chippa United
5	Golden Arrows
6	Kaizer Chiefs
7	Mamelodi Sundowns
8	Moroka Swallows
9	Orlando Pirates

QUESTION 4:

```
49      -- 4. Count how many players are in each team
50      SELECT COUNT(Player_name) AS Number_of_Players,
51             Team
52      FROM [SQLFundamentals].[dbo].[ketro_sa_soccer_dataset_advanced]
53      GROUP BY Team;
```

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Results Messages

	Number_of_Players	Team
1	22	AmaZulu FC
2	17	Bloemfontein Celtic
3	15	Cape Town City
4	21	Chippa United
5	23	Golden Arrows
6	20	Kaizer Chiefs

QUESTION 5:

```
55      -- 5. Identify the top 10 players with the most goals.
56      SELECT Top (10) Goals,
57             Player_name
58      FROM [SQLFundamentals].[dbo].[ketro_sa_soccer_dataset_advanced]
59      ORDER BY Goals DESC;
```

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Results Messages

	Goals	Player_name
1	99	Vusi Molefe
2	98	Thabo Ndlovu
3	98	Gugu Hlongwane
4	98	Thabo Sithole
5	98	Mandla Mabena
6	97	Mpho Mahlangu
7	96	Boitumelo Nkosi
8	92	Mpho Radebe
9	92	Khanyi Nkosi
10	91	Sipho Phiri

QUESTION 6:

```

61      -- 6. Find the average salary for players in each team.
62      SELECT Team,
63              AVG(CAST(average_salary_zar AS FLOAT)) AS avg_salary
64      FROM [SQLFundamentals].[dbo].[ketros_soccer_dataset_advanced]
65      GROUP BY team;
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	Team	avg_salary
1	AmaZulu FC	172467.569545455
2	Bloemfontein Celtic	163683.838823529
3	Cape Town City	208407.432
4	Chippa United	180947.634761905
5	Golden Arrows	199057.643913043
6	Kaizer Chiefs	188954.4615
7	Mamelodi Sundowns	194404.484736842
8	Moroka Swallows	186786.827894737
9	Orlando Pirates	178570.593333333
10	Polokwane City	171738.877777778
11	Richards Bay FC	193006.199230769

QUESTION 7:

```

66      -- 7. Retrieve the top 10 players with the highest market value
67
68      SELECT TOP (10) Player_name,
69                      Market_value_zar
70      FROM [SQLFundamentals].[dbo].[ketros_soccer_dataset_advanced]
71      ORDER BY Market_value_zar DESC;
72

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	Player_name	Market_value_zar
1	Nokuthula Baloyi	9906925.35
2	Gugu Mokoena	9868061.36
3	Khanyi Mthembu	9845842.09
4	Nomsa Hlongwane	975559.35
5	Kabelo Mahlangu	9606957.95
6	Kagiso Dlamini	9383118.97
7	Lindiwe Phiri	9352111.61
8	Mpho Radebe	9294272.32
9	Lindiwe Molefe	9277927.61
10	Thembi Tshabalala	9091048.14

QUESTION 8:

```

73  -- 8. Calculate the average passing accuracy for each position.
74  SELECT Position,
75         AVG(CAST(Passing_accuracy AS FLOAT)) AS Average_passing_accuracy
76  FROM [SQLFundamentals].[dbo].[ketro_sa_soccer_dataset_advanced]
77  GROUP BY Position;
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79
80
81
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Results Messages

	Position	Average_passing_accuracy
1	Defender	82,6642857142857
2	Forward	83,210843373494
3	Goalkeeper	83,9397058823529
4	Midfielder	82,8177215189873

QUESTION 9:

```

79  -- 9. Compare shot accuracy with goals to find correlations.
80  SELECT Goals,
81         AVG(CAST(Shot_accuracy AS FLOAT)) AS Average_shot_accuracy,
82         SUM(Goals*CAST(Shot_accuracy AS FLOAT)) AS Goal_and_shot_correlation
83  FROM [SQLFundamentals].[dbo].[ketro_sa_soccer_dataset_advanced]
84  GROUP BY Goals
85  ORDER BY Goals DESC;
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87
88
89
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91
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Results Messages

	Goals	Average_shot_accuracy	Goal_and_shot_correlation
1	99	79	7821
2	98	71.75	28126
3	97	80.8	7837.6
4	96	78.3	7516.8
5	92	70.3	12935.2
6	91	75.3	34261.5
7	90	76.6666666666667	20700
8	9	62.3928571428571	7861.5
9	89	56.8	10110.4
10	88	73.8	6494.4

QUESTION 10:

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86
87 -- 10. Compute total goals and assists for each team.
88 SELECT Team,
89         SUM(CAST(Goals AS FLOAT)) AS Number_of_Goals,
90         SUM(CAST(Assists AS FLOAT)) AS Number_of_Assists
91 FROM [SQLFundamentals].[dbo].[ketrosoccerdatasetadvanced]
92 GROUP BY Team
93 ORDER BY Number_of_Goals;
94
95
96

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	Team	Number_of_Goals	Number_of_Assists
1	Orlando Pirates	170	225
2	Cape Town City	374	284
3	Richards Bay FC	453	273
4	TS Galaxy	541	337
5	Royal AM	544	480
6	AmaZulu FC	546	568
7	Mamelodi Sundowns	570	475
8	SuperSport United	580	514
9	Kaizer Chiefs	597	481
10	Bloemfontein Celtic	613	566

QUESTION 11:

```

94
95 -- 11. Count players by their marital status
96 SELECT Marital_status,
97         Team,
98         COUNT(*) AS Number_of_players
99 FROM [SQLFundamentals].[dbo].[ketrosoccerdatasetadvanced]
100 GROUP BY Marital_status, Team;
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	Marital_status	Team	Number_of_players
1	Divorced	AmaZulu FC	10
2	Married	AmaZulu FC	4
3	Single	AmaZulu FC	4
4	Widowed	AmaZulu FC	4
5	Divorced	Bloemfontein Celtic	5
6	Married	Bloemfontein Celtic	3
7	Single	Bloemfontein Celtic	3
8	Widowed	Bloemfontein Celtic	6
9	Divorced	Cape Town City	3
10	Married	Cape Town City	5

QUESTION 12:

```

101
102 -- 12. Count players by nationality.
103 SELECT Nationality,
104         Team,
105         COUNT(*) Number_of_players
106 FROM [SQLFundamentals].[dbo].[ketros_sa_soccer_dataset_advanced]
107 GROUP BY Nationality, Team;
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109
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Results Messages

	Nationality	Team	Number_of_players
1	Ghanaian	AmaZulu FC	3
2	Malawian	AmaZulu FC	5
3	Mozambican	AmaZulu FC	3
4	Nigerian	AmaZulu FC	1
5	South African	AmaZulu FC	3
6	Zambian	AmaZulu FC	2
7	Zimbabwean	AmaZulu FC	5
8	Ghanaian	Bloemfontein Celtic	1
9	Mozambican	Bloemfontein Celtic	3
10	South African	Bloemfontein Celtic	5
11	Zambian	Bloemfontein Celtic	7

QUESTION 13:

```

109 -- 13. Find average market value grouped by nationality.
110 SELECT Nationality,
111         AVG(CAST(Market_value_zar AS FLOAT)) AS Average_market_value
112 FROM [SQLFundamentals].[dbo].[ketros_sa_soccer_dataset_advanced]
113 GROUP BY Nationality
114 ORDER BY Average_market_value DESC;
115
116
117
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Results Messages

	Nationality	Average_market_value
1	Nigerian	15069261.3964103
2	Mozambican	14736203.5943243
3	Zambian	12777043.0611364
4	Ghanaian	12298308.0365957
5	South African	12037870.2363043
6	Malawian	11727296.0554762
7	Zimbabwean	10367909.5322222

QUESTION 14:

```
115
116
117 -- 14. Determine how many player contracts end in each year.
118 SELECT Contract_end_year,
119         COUNT(*) AS Number_of_Players
120 FROM [SQLFundamentals].[dbo].[ketro_sa_soccer_dataset_advanced]
121 GROUP BY Contract_end_year;
122
123
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125
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```

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Results Messages

	Contract_end_year	Number_of_Players
1	2026	63
2	2027	70
3	2028	52
4	2029	50
5	2030	65

QUESTION 15:

```
121
122
123 -- 15. Identify players whose contracts end next year.
124 SELECT Contract_end_year,
125         COUNT(*) AS Number_of_Players
126 FROM [SQLFundamentals].[dbo].[ketro_sa_soccer_dataset_advanced]
127 GROUP BY Contract_end_year
128 HAVING Contract_end_year = 2026;
129
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131
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```

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Results Messages

	Contract_end_year	Number_of_Players
1	2026	63

QUESTION 16:

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128
129 -- 16. Summarize the number of players by injury status
130 SELECT Injury_status,
131        COUNT(*) AS Number_of_players
132 FROM [SQLFundamentals].[dbo].[ketrosoccerdataset_advanced]
133 GROUP BY Injury_status;
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135
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137
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139
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141

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Results Messages

	Injury_status	Number_of_players
1	Healthy	99
2	Injured	97
3	Recovering	104

QUESTION 17:

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134
135 -- 17. Calculate goals per match ratio for each player.
136 SELECT Team,
137        Player_name,
138        Matches_played,
139        Goals,
140        CAST(Goals AS Float)/NULLIF(Matches_played, 0)*100 AS Goals_and_Match_Ratio
141 FROM [SQLFundamentals].[dbo].[ketrosoccerdataset_advanced];
142

```

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Results Messages

	Team	Player_name	Matches_played	Goals	Goals_and_Match_Ratio
	Stellenbosch FC	Siyanda Dlamini	53	9	16.9811320754717
	Cape Town City	Thabo Ndlovu	272	98	36.0294117647059
	Stellenbosch FC	Vusi Molefe	398	99	24.8743718592965
	Bloemfontein Celtic	Thembi Mahlangu	135	1	0.740740740740741
	Polokwane City	Nokuthula Sithole	116	2	1.72413793103448
	Kaizer Chiefs	Thembi Sithole	279	3	1.0752688172043
	Chippa United	Siyanda Mahlangu	18	6	33.3333333333333
	Polokwane City	Lerato Mashaba	241	77	31.9502074688797
	Polokwane City	Nomsa Mahlangu	322	80	24.8447204968944
0	Kaizer Chiefs	Tumelo Khumalo	252	48	19.047619047619
1	TS Galaxy	Gugu Molefe	379	7	1.84696569920844

QUESTION 18:

```

142
143 -- 18. Count how many players are managed by each agent.
144 SELECT Agent,
145         COUNT(*) AS Number_of_Player
146 FROM [SQLFundamentals].[dbo].[ketrosoccerdataset_advanced]
147 GROUP BY Agent;
148
149
150
151
152
153
154
155

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100 % No issues found

Results Messages

	Agent	Number_of_Player
1	None	62
2	PlayerFirst	63
3	ProSport	62
4	SA Elite Agents	51
5	SoccerLink Africa	62

QUESTION 19:

```

149 -- 19. Calculate average height and weight by player position
150 SELECT Position,
151         Player_name,
152         AVG(CAST(Height_cm AS FLOAT)) AS Player_height,
153         AVG(CAST(Weight_kg AS FLOAT)) AS Player_weight
154 FROM [SQLFundamentals].[dbo].[ketrosoccerdataset_advanced]
155 GROUP BY Position, Player_name;
156
157
158
159
160
161

```

100 % No issues found

Results Messages

	Position	Player_name	Player_height	Player_weight
1	Goalkeeper	Ayanda Khumalo	165	68
2	Midfielder	Ayanda Khumalo	171	63
3	Defender	Ayanda Mabaso	169	90
4	Goalkeeper	Ayanda Mabaso	189	90
5	Midfielder	Ayanda Mokoena	169	92
6	Forward	Ayanda Molefe	184.5	74
7	Defender	Ayanda Mthembu	179	85
8	Forward	Ayanda Sithole	179	60
9	Midfielder	Ayanda Sithole	172	72
10	Goalkeeper	Ayanda Zulu	190	92
11	Midfielder	Boitumelo Mabaso	184	92

QUESTION 20:

```

157 -- 20. Identify players with the highest combined goals and assists.
158 SELECT TOP 10
159     Player_name,
160     Team,
161     Goals,
162     Assists,
163     SUM(CAST(Goals + Assists AS FLOAT)) Goals_and_Assists
164 FROM [SQLFundamentals].[dbo].[ketro_sa_soccer_dataset_advanced]
165 GROUP BY Player_name, Team, Goals, Assists
166 ORDER BY Goals_and_Assists DESC;
167
168
169

```

100 % No issues found

Results Messages

	Player_name	Team	Goals	Assists	Goals_and_Assists
1	Siyanda Mabena	Sekhukhune United	118	75	11875
2	Zanele Phiri	TS Galaxy	117	44	11744
3	Kagiso Phiri	Richards Bay FC	117	25	11725
4	Zanele Molefe	SuperSport United	116	76	11676
5	Nokuthula Mabena	SuperSport United	114	53	11453
6	Karabo Phiri	Chippa United	114	29	11429
7	Nokuthula Mthembu	Mamelodi Sundowns	114	21	11421
8	Lindiwe Radebe	Chippa United	113	41	11341
9	Vusi Mabena	Richards Bay FC	113	18	11318
10	Vusi Radebe	Stellenbosch FC	112	76	11276